

6 – 3 – 2026, SSC Algebra Board Exam Solution

Time: 2 Hours

Mathematics (Part – I)

Max. Marks: 40

1. (A) For every subquestion, four alternative answers are given. Choose the correct alternative [4] and write the alphabet :

(i) Rate of GST on essential commodities is

(A) 5%

(B) 12%

(C) 0%

(D) 18%

Ans: (C) 0%

(ii) Out of the following equations which one is not a quadratic equation ?

(A) $x^2 + 4x = 11 + x^2$ (B) $x^2 = 4x$ (C) $5x^2 = 90$ (D) $2x - x^2 = x^2 + 5$

Ans: (A) $x^2 + 4x = 11 + x^2$

EXPLANATION:

$$x^2 + 4x = 11 + x^2 \quad \therefore 4x = 11$$

Here x^2 term gets cancelled and the equation reduces to degree 1. $\therefore$ It is not a quadratic equation.

(iii) To draw graph of $4x + 5y = 19$, find the value of y when $x = 1$.

(A) 4

(B) 3

(C) 2

(D) -3

Ans: (B) 3

EXPLANATION:

$$4x + 5y = 19$$

$$\text{Put } x = 1$$

$$\therefore 4 \times 1 + 5y = 19 \quad \therefore 5y = 19 - 4 = 15$$

$$\therefore y = \frac{15}{5} = 3$$

(iv) Sum of first five multiples of 3 is

(A) 45

(B) 55

(C) 15

(D) 75

Ans: (A) 45

EXPLANATION:

First five multiples of 3 form an A.P. : 3, 6, 9, 12, 15

$$a = t_1 = 3, t_5 = 15, n = 5, S_5 = ?$$

$$S_n = \frac{n}{2} [a + t_n] \text{ (Formula)}$$

$$S_5 = \frac{5}{2} [3 + 15] = \frac{5}{2} \times 18 = 5 \times 9 = 45$$

$$\text{Or, simply: } 3 + 6 + 9 + 12 + 15 = 45$$

(B) Solve the following subquestions:

[4]

(i) On certain article if rate of CGST is 9%, then what is the rate of GST ?

Ans:

$$\text{CGST} = 9\%$$

$$\text{We know that, } \text{CGST} = \frac{1}{2} \times \text{GST}$$

$$\therefore \text{GST} = 2 \times \text{CGST} = 2 \times 9\%$$

$$\therefore \text{GST} = \mathbf{18\%}$$

(ii) For simultaneous equations in variables x and y , if $D_x = -42$ and $D = -3$, then find the value of x .

Ans:

$$D_x = -42, D = -3, x = ?$$

$$x = \frac{D_x}{D} = \frac{-42}{-3}$$

$$\therefore x = \mathbf{14}$$

(iii) When two coins are tossed simultaneously, write the sample space 'S'.

Ans:

Two coins are tossed simultaneously.

∴ The sample space is

$$S = \{ HH, HT, TH, TT \}$$

(iv) Write the following quadratic equation in the form $ax^2 + bx + c = 0$:

$$2y = 10 - y^2$$

Ans:

$$2y = 10 - y^2$$

∴ $y^2 + 2y - 10 = 0$ is the required form $ax^2 + bx + c = 0$,

where $a = 1, b = 2, c = -10$

2. (A) Complete and write any two activities from the following:

[4]

(i) If one die is rolled, then find the probability of the event, that the number on the upper face is prime.

Activity:

'S' is the sample space.

$$S = \{ 1, 2, 3, 4, 5, 6 \}$$

$$\therefore n(S) = \boxed{}$$

Event A : Prime number on the upper face.

$$\therefore A = \{ 2, 3, 5 \}$$

$$\therefore n(A) = \boxed{}$$

$$\therefore P(A) = \frac{n(A)}{\boxed{}} \text{ (formula)}$$

$$\therefore P(A) = \frac{\boxed{}}{6}$$

$$= \frac{1}{2}$$

Ans:

'S' is the sample space.

$$S = \{ 1, 2, 3, 4, 5, 6 \}$$

$$\therefore n(S) = \boxed{6}$$

Event A : Prime number on the upper face.

$$\therefore A = \{ 2, 3, 5 \}$$

$$\therefore n(A) = \boxed{3}$$

$$\therefore P(A) = \frac{n(A)}{\boxed{n(S)}} \text{ (formula)}$$

$$\therefore P(A) = \frac{\boxed{3}}{6}$$

$$\therefore P(A) = \frac{1}{2}$$

(ii) Write the correct number in the given boxes from the given A.P. : 70, 60, 50, 40,

$$\text{Here, } t_1 = \boxed{}, t_2 = 60, t_3 = \boxed{}$$

$$\therefore a = \boxed{}, d = \boxed{}$$

Ans:

$$\text{Here, } t_1 = \boxed{70}, t_2 = 60, t_3 = \boxed{50}$$

First term $a = t_1 = 70$ and common difference $d = t_2 - t_1 = 60 - 70 = -10$

$$\therefore a = \boxed{70}, d = \boxed{-10}$$

(iii) Complete the following table to draw graph of the equation $x + y = 3$:

Activity:

x	3	
y		5
(x, y)		

Ans:

x	3	-2
y	0	5
(x, y)	(3, 0)	(-2, 5)

EXPLANATION:

Put $x = 3$ in $x + y = 3$, we get

$$3 + y = 3 \quad \therefore y = 0$$

Put $y = 5$ in $x + y = 3$, we get

$$x + 5 = 3 \quad \therefore x = -2$$

(B) Solve any four subquestions from the following:

[8]

(i) If two coins are tossed simultaneously, find the probability of the event, "getting at least one head."

Ans:

Two coins are tossed simultaneously.

Let 'S' be the sample space.

$$\therefore S = \{ HH, HT, TH, TT \}, \quad \therefore n(S) = 4$$

Let Event A : getting at least one head.

$$\therefore A = \{ HH, HT, TH \}, \quad \therefore n(A) = 3$$

$$\therefore P(A) = \frac{n(A)}{n(S)} \quad \dots \text{(formula)}$$

$$\therefore P(A) = \frac{3}{4}$$

Probability of getting at least one head is $\frac{3}{4}$.

(ii) Solve the following simultaneous equations :

$$x + y = 4 ; \quad 2x - y = 2$$

Ans:

$$x + y = 4 \dots\dots\dots (I)$$

$$2x - y = 2 \dots\dots\dots (II)$$

Add equation (I) and (II),

$$\begin{array}{rcl} x + y & = & 4 \\ + 2x - y & = & 2 \\ \hline 3x & = & 6 \end{array}$$

$$\therefore x = \frac{6}{3} \quad \therefore \mathbf{x = 2}$$

Put $x = 2$ in equation (I), we get,

$$2 + y = 4$$

$$\therefore y = 4 - 2 \quad \therefore \mathbf{y = 2}$$

$x = 2$ and $y = 2$ is the solution.

(iii) Solve the following quadratic equation by factorisation method :

$$x^2 + x - 20 = 0$$

Ans:

$$x^2 + x - 20 = 0$$

$$x^2 + 5x - 4x - 20 = 0 \quad \dots\dots [5 \times (-4) = -20 ; 5 - 4 = 1]$$

$$\therefore x(x + 5) - 4(x + 5) = 0$$

$$\therefore (x + 5)(x - 4) = 0$$

$$\therefore x + 5 = 0 \quad \text{or} \quad x - 4 = 0$$

$$\therefore \mathbf{x = -5} \quad \text{or} \quad \mathbf{x = 4}$$

The roots of the given quadratic equation are -5 and 4 .

(iv) Given A.P. is 12, 16, 20, 24,

Find the 24th term of this progression.

Ans:

Given A.P. : 12, 16, 20, 24,

$$a = t_1 = 12, t_2 = 16, t_{24} = ?$$

$$d = t_2 - t_1 = 16 - 12 = 4$$

$$t_n = a + (n - 1)d \quad \dots\dots \text{(Formula)}$$

$$\therefore t_{24} = 12 + (24 - 1) \times 4$$

$$\therefore t_{24} = 12 + 23 \times 4 = 12 + 92$$

$$\therefore t_{24} = \mathbf{104}$$

24th term of the given A.P. is 104.

(v) If $\sum x_i f_i = 1265$ and $N = 50$, then find the mean of the given data.

Ans:

$$\sum x_i f_i = 1265, N = 50, \bar{X} = ?$$

$$\text{Mean, } \bar{X} = \frac{\sum x_i f_i}{N} \quad \dots \text{(formula)}$$

$$\therefore \bar{X} = \frac{1265}{50}$$

$$\therefore \text{Mean, } \bar{X} = \mathbf{25.3}$$

3. (A) Complete and write any one activity from the following:

[3]

(i) Shreekar bought a Laptop with 10% discount on printed price. The printed price of that Laptop was ₹ 50,000. 18% GST was charged on discounted price. Find the amount of CGST and SGST. What amount did Shreekar pay ?

Solution :

$$\text{Discount} = 10\% \text{ of } 50,000 = ₹ \boxed{}$$

$$\therefore \text{Taxable value of Laptop} = 50,000 - \boxed{}$$

$$= ₹ 45,000$$

$$\therefore \text{Rate of GST} = \boxed{}$$

$$\therefore \text{Rate of CGST} = 9\%$$

$$\therefore \text{CGST} = 9\% \text{ of } 45,000$$

$$= \boxed{} \times 45,000$$

$$= ₹ \boxed{}$$

$$\therefore \text{SGST} = ₹ 4,050$$

$$\text{Total amount of Laptop} = 45,000 + 4,050 + 4,050 = ₹ \boxed{}$$

Ans:

$$\text{Discount} = 10\% \text{ of } 50,000 = ₹ \boxed{5,000}$$

$$\therefore \text{Taxable value of Laptop} = 50,000 - \boxed{5,000}$$

$$= ₹ 45,000$$

$$\therefore \text{Rate of GST} = \boxed{18\%}$$

$$\therefore \text{Rate of CGST} = 9\%$$

$$\therefore \text{CGST} = 9\% \text{ of } 45,000$$

$$= \boxed{\frac{9}{100}} \times 45,000$$

$$= ₹ \boxed{4,050}$$

$$\therefore \text{SGST} = ₹ 4,050$$

$$\text{Total amount of Laptop} = 45,000 + 4,050 + 4,050 = ₹ \boxed{53,100}$$

(ii) One of the roots of the quadratic equation $5m^2 + 2m + k = 0$ is $-\frac{7}{5}$. Complete the following activity to find the value of ' k '.

Solution :

$-\frac{7}{5}$ is a root of quadratic equation $5m^2 + 2m + k = 0$.

$\therefore$ put $m = -\frac{7}{5}$ in the equation.

$$\therefore 5 \times \boxed{}^2 + 2 \times \boxed{} + k = 0$$

$$\therefore \boxed{} + \boxed{} + k = 0$$

$$\therefore \boxed{} + k = 0$$

$$\therefore k = \boxed{}$$

Ans:

$-\frac{7}{5}$ is a root of quadratic equation $5m^2 + 2m + k = 0$.

$\therefore$ put $m = -\frac{7}{5}$ in the equation.

$$\therefore 5 \times \boxed{\frac{-7}{5}}^2 + 2 \times \boxed{\frac{-7}{5}} + k = 0$$

$$\therefore \boxed{\frac{49}{5}} + \boxed{\frac{-14}{5}} + k = 0$$

$$\therefore \boxed{7} + k = 0$$

$$\therefore k = \boxed{-7}$$

(B) Solve any two subquestions of the following :

[6]

(i) Two numbers differ by 3. The sum of twice the smaller number and thrice the greater number is 19. Find the numbers.

Ans:

Let the greater number be 'x' and the smaller number be 'y'.

From the first given condition,

$$x - y = 3 \text{ (I)}$$

From the second given condition,

$$3x + 2y = 19 \text{ (II)}$$

Multiplying equation (I) by 2,

$$2x - 2y = 6 \text{ (III)}$$

Adding equation (II) and (III),

$$\begin{array}{rcl} 3x + 2y & = & 19 \\ + 2x - 2y & = & 6 \\ \hline 5x & = & 25 \end{array}$$

$$\therefore x = \frac{25}{5} = 5$$

Put $x = 5$ in equation (I), we get,

$$5 - y = 3$$

$$\therefore -y = 3 - 5 = -2 \quad \therefore y = 2$$

5 and 2 are the two required numbers.

(ii) Sachin invested some amount in a National Saving Certificate Scheme. In the first year he invested ₹ 5,000, in the second year ₹ 7,000, in the third year ₹ 9,000 and so on. Find the total amount that he invested in 12 years.

Ans:

The amounts invested by Sachin in consecutive years form an A.P. :

5000, 7000, 9000,

Here, $a = 5000$, $d = 7000 - 5000 = 2000$, $n = 12$, $S_{12} = ?$

$$S_n = \frac{n}{2} [2a + (n - 1)d] \text{ (formula)}$$

$$\therefore S_{12} = \frac{12}{2} [2 \times 5000 + (12 - 1) \times 2000]$$

$$= 6 \times [10000 + 22000]$$

$$= 6 \times 32000$$

$$\therefore S_{12} = 192000$$

In 12 years, Sachin invested ₹ 1,92,000.

(iii) There are six cards in a box, each bearing a number from 0 to 5. Find the probability of each of the following events, that :

(1) The number on the card drawn is a natural number.

(2) The number on the card drawn is a whole number.

Ans:

Let 'S' be the sample space.

$$\therefore S = \{ 0, 1, 2, 3, 4, 5 \}, \quad \therefore n(S) = 6$$

(1) Let Event A : The number on the card drawn is a natural number.

$$\therefore A = \{ 1, 2, 3, 4, 5 \}, \quad \therefore n(A) = 5$$

$$\therefore P(A) = \frac{n(A)}{n(S)} \quad \dots \text{(formula)}$$

$$\therefore P(A) = \frac{5}{6}$$

(2) Let Event B : The number on the card drawn is a whole number.

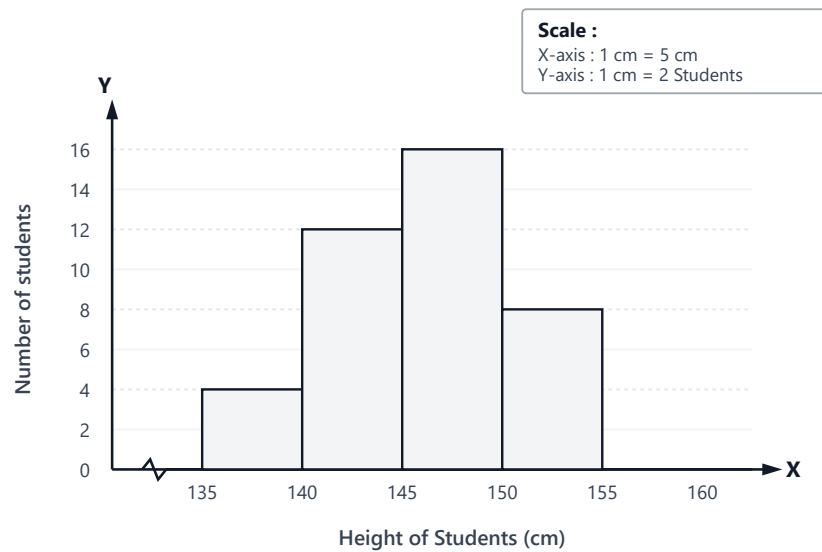
$$\therefore B = \{ 0, 1, 2, 3, 4, 5 \}, \quad \therefore n(B) = 6$$

$$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{6}{6} = \mathbf{1}$$

(iv) Draw a histogram of the following data :

Height of Students (cm)	No. of Students
135 — 140	4
140 — 145	12
145 — 150	16
150 — 155	8

Ans:



4. Solve any two subquestions of the following :

[8]

(i) The sum of two roots of a quadratic equation is 7 and sum of their cubes is 91, find the equation.

Ans:

Let the two roots be α and β .

According to the given conditions,

$$\alpha + \beta = 7 \quad \text{and} \quad \alpha^3 + \beta^3 = 91 \quad \dots (i)$$

We know that,

$$(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta) \quad \dots \text{[formula]}$$

$$\therefore (7)^3 = 91 + 3\alpha\beta(7) \quad \dots \text{[from (i)]}$$

$$\therefore 343 = 91 + 21\alpha\beta$$

$$\therefore 343 - 91 = 21\alpha\beta$$

$$\therefore 252 = 21\alpha\beta$$

$$\therefore \alpha\beta = \frac{252}{21} = 12 \quad \dots (ii)$$

Quadratic Equation is given by,

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\therefore x^2 - 7x + 12 = 0 \quad \dots \text{[from (i) and (ii)]}$$

The required quadratic equation is **$x^2 - 7x + 12 = 0$** .

(ii) Smt. Tejaswini purchased 100 shares of FV ₹ 80 when the MV is ₹ 900. She paid brokerage at the rate of 0.5% and 18% GST on brokerage. Find the following :

- (1) Net amount paid for 100 shares.
- (2) Brokerage paid on sum invested.
- (3) GST paid on brokerage.
- (4) Total amount paid for 100 shares.

Ans:

FV = ₹ 80, MV = ₹ 900, Number of shares = 100,

Rate of brokerage = 0.5%, Rate of GST on brokerage = 18%

(1) Net amount paid for 100 shares = MV × Number of shares
= $900 \times 100 = \text{₹ } 90,000$

(2) Brokerage = 0.5% of sum invested = $\frac{0.5}{100} \times 90000$
= **₹ 450**

(3) GST on brokerage = 18% of 450 = $\frac{18}{100} \times 450$
= **₹ 81**

(4) Total amount paid for 100 shares = $90,000 + 450 + 81$
= **₹ 90,531**

(iii) The following table shows the funds collected by 105 students for flood affected people. Find the median of the data.

Fund (Rupees)	No. of Students
10 — 20	10
20 — 30	37
30 — 40	15
40 — 50	20
50 — 60	7
60 — 70	8
70 — 80	8

Ans:

Class (Fund in ₹)	Frequency (No. of students)	Cumulative frequency (less than upper limit)
10 — 20	10	10
20 — 30	37	47 $\rightarrow cf$
30 — 40	15 $\rightarrow f$	62
40 — 50	20	82
50 — 60	7	89
60 — 70	8	97
70 — 80	8	105

$$N = 105, \quad \frac{N}{2} = \frac{105}{2} = 52.5$$

Cumulative frequency which is just greater than 52.5 is 62.

$\therefore$ The median class is 30 — 40.

$$L = 30, cf = 47, f = 15, h = 10$$

$$\text{Median} = L + \frac{\frac{N}{2} - cf}{f} \times h \quad \dots \text{(formula)}$$

$$\therefore = 30 + \frac{52.5 - 47}{15} \times 10$$

$$\therefore = 30 + \frac{5.5 \times 10}{15} = 30 + \frac{55}{15} = 30 + 3.67$$

Median = **33.67**

∴ The median of the funds collected is ₹ 33.67

5. Solve any one subquestion from the following :

[3]

(i) A survey was conducted in the school. Student using different modes of transport is given below :

Mode of Transport	Percentage of Students
Bicycle	35
Bus	a
Railway	$2a$
Rickshaw	20

Find the value of a and draw the Pie diagram.

Ans:

The total of all the percentages is 100.

$$\therefore 35 + a + 2a + 20 = 100$$

$$\therefore 3a + 55 = 100$$

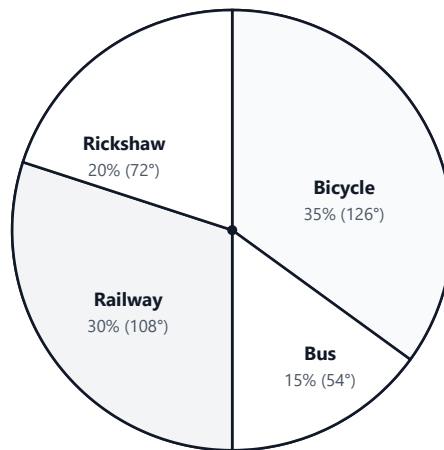
$$\therefore 3a = 100 - 55 = 45$$

$$\therefore a = \frac{45}{3} = \mathbf{15}$$

$$\therefore \text{Bus} = a = 15\% \text{ and } \text{Railway} = 2a = 30\%$$

Measures of central angles :

Mode of Transport	Percentage of Students	Measure of central angle
Bicycle	35	$\frac{35}{100} \times 360^\circ = \mathbf{126^\circ}$
Bus	15	$\frac{15}{100} \times 360^\circ = \mathbf{54^\circ}$
Railway	30	$\frac{30}{100} \times 360^\circ = \mathbf{108^\circ}$
Rickshaw	20	$\frac{20}{100} \times 360^\circ = \mathbf{72^\circ}$
Total	100	360°



(ii) There are some students in group A and group B. If one student goes from group A to group B, then the number of students in both groups is same. But one student goes from group B to group A, then the number of students in group A becomes twice of group B. Find the number of students in each group.

Ans:

Let the number of students in group A be x and in group B be y .

When one student goes from group A to group B, both groups have the same number of students.

$$\therefore x - 1 = y + 1$$

$$\therefore x - y = 2 \text{ (I)}$$

When one student goes from group B to group A, group A becomes twice of group B.

$$\therefore x + 1 = 2(y - 1)$$

$$\therefore x + 1 = 2y - 2$$

$$\therefore x - 2y = -3 \text{ (II)}$$

Subtracting equation (II) from equation (I),

$$x - y = 2$$

$$x - 2y = -3$$

$$\begin{array}{r} (-) \quad (+) \quad (+) \\ \hline \end{array}$$

$$y = 5$$

Put $y = 5$ in equation (I), we get,

$$x - 5 = 2 \quad \therefore x = 7$$

There are **7 students in group A** and **5 students in group B**.